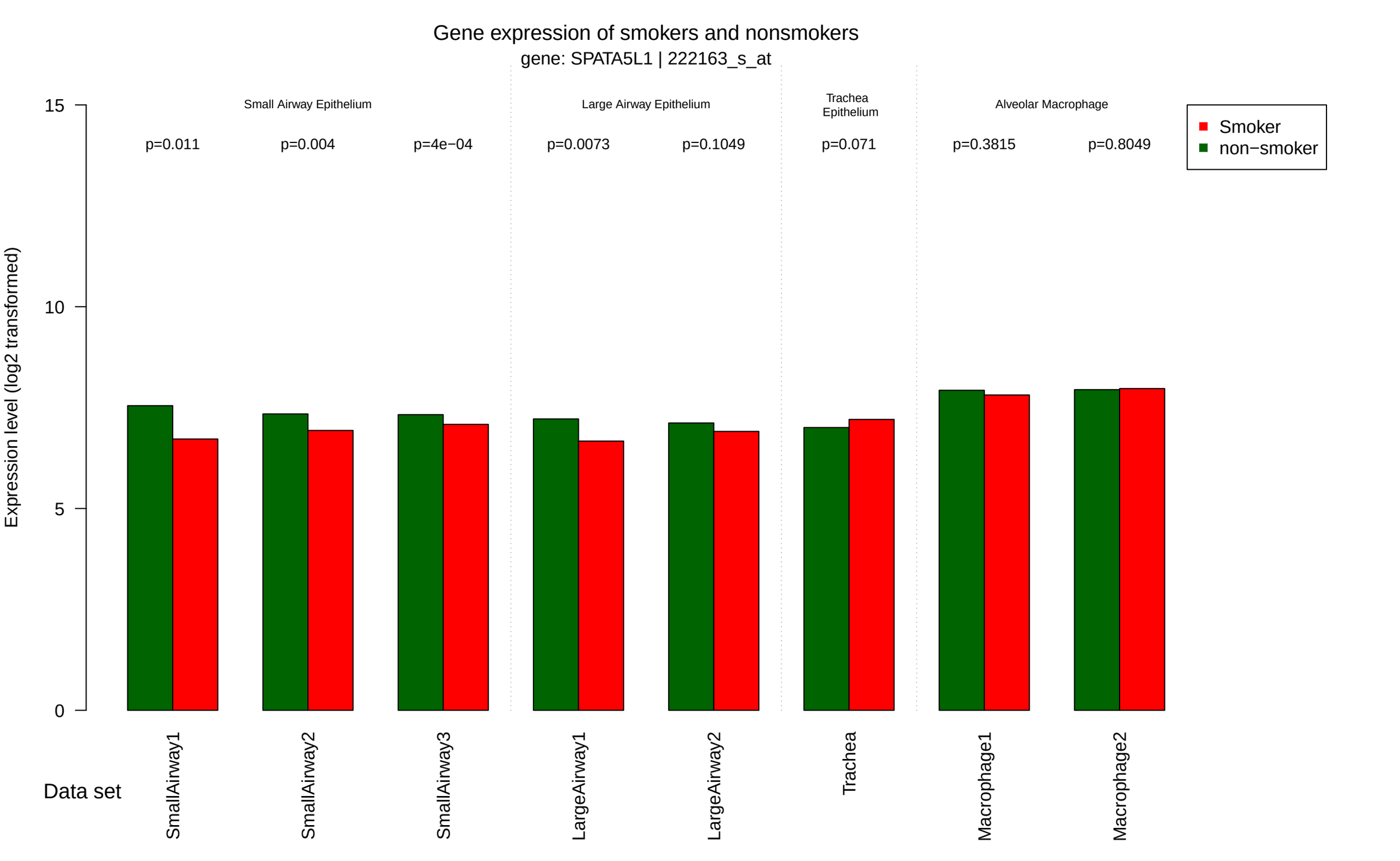
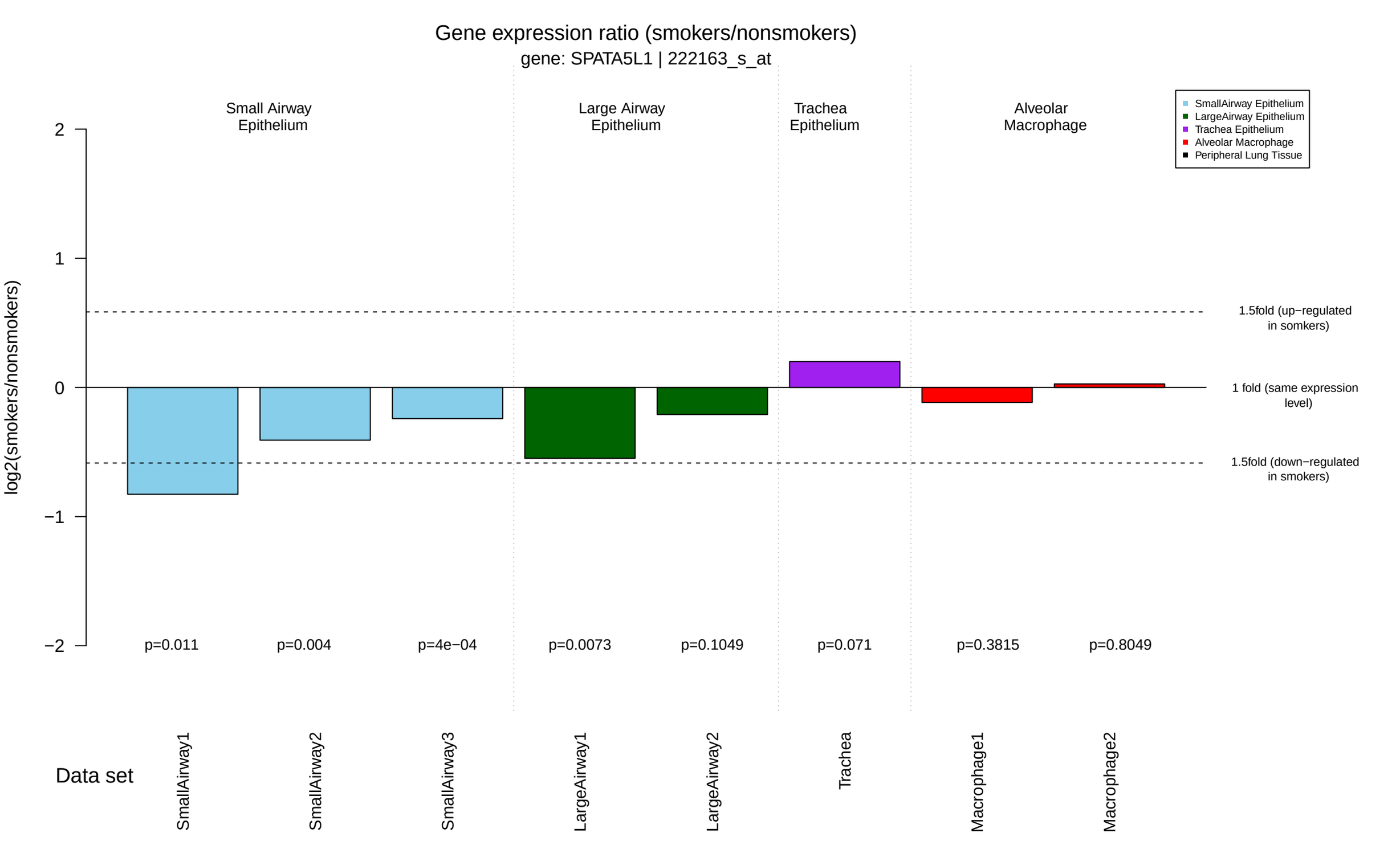
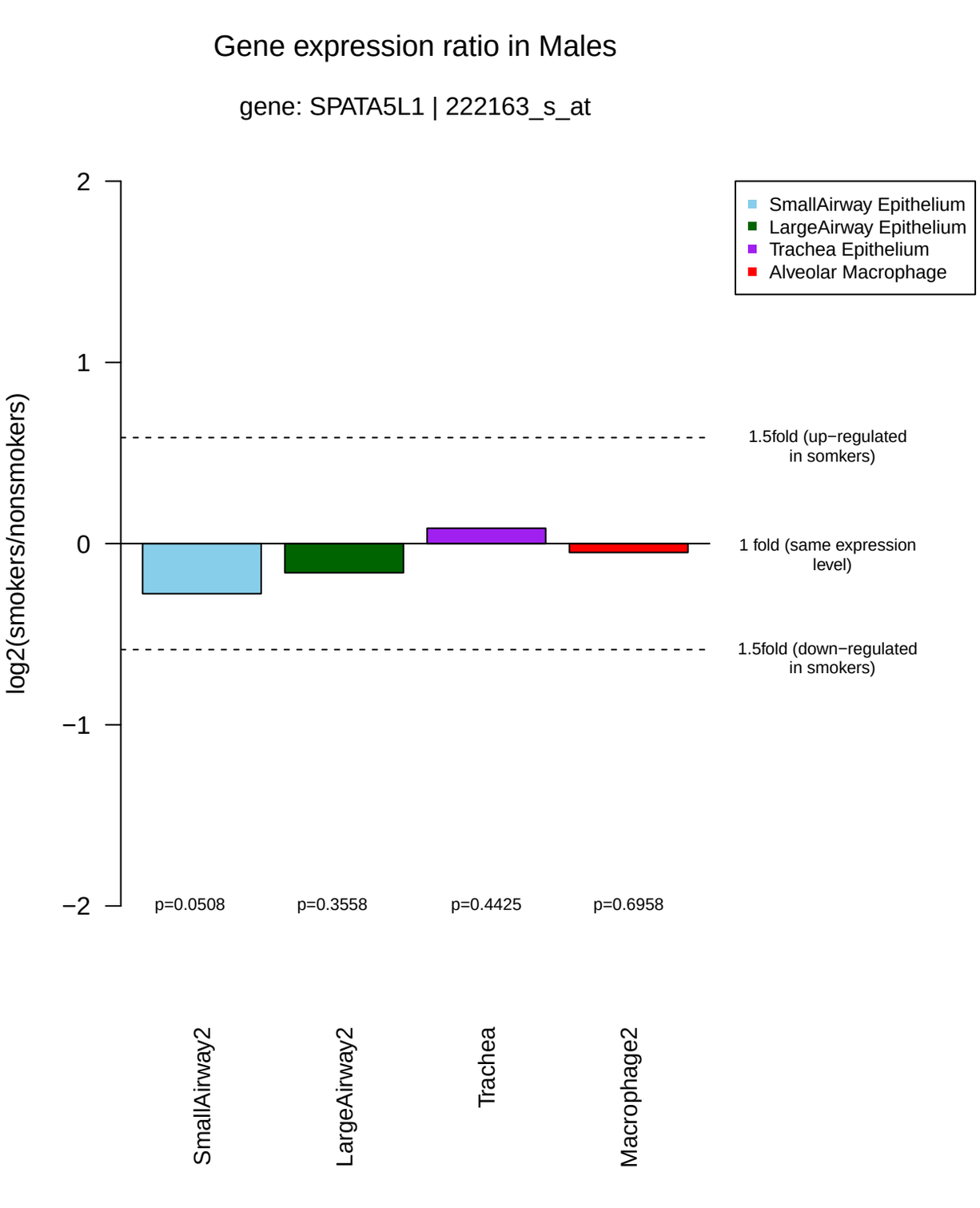
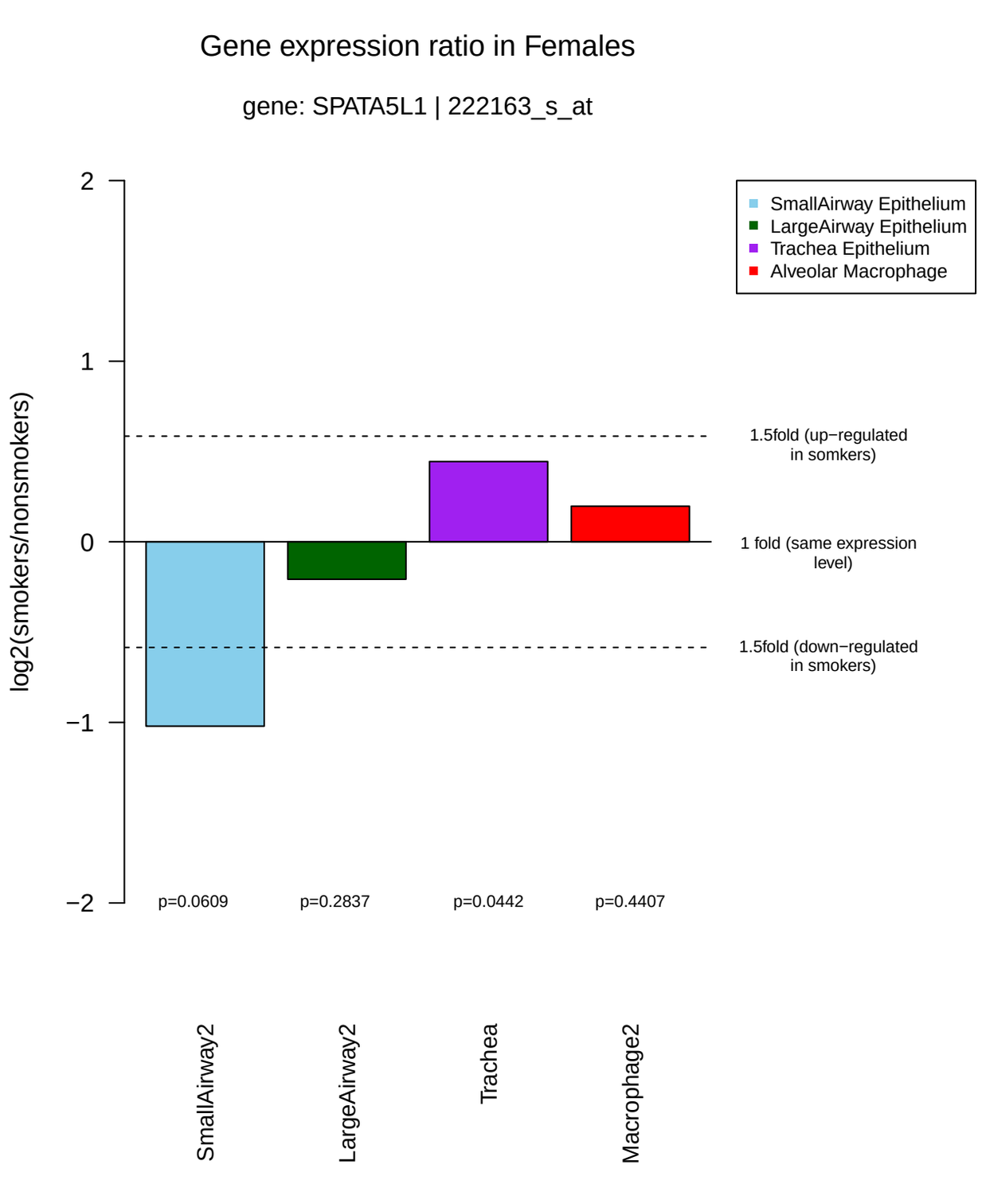
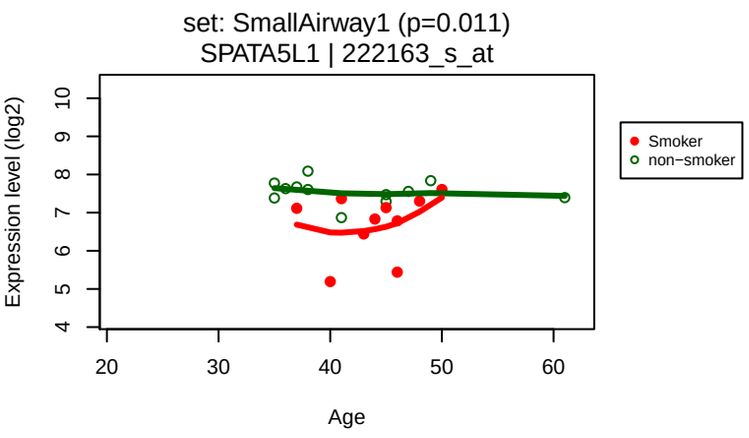
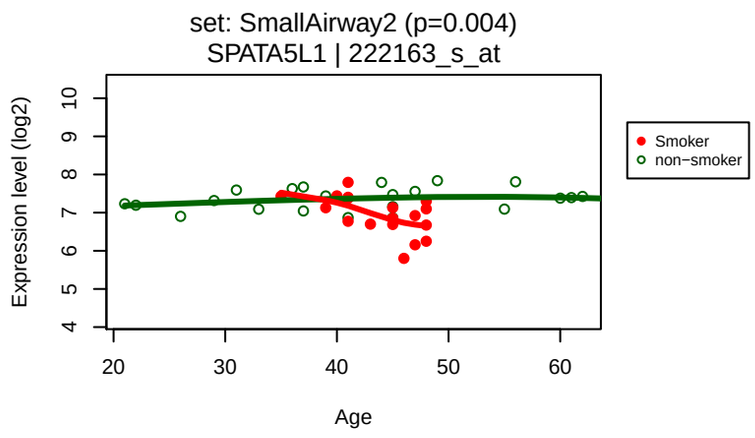

Gene expression of smokers and nonsmokers
gene: SPATA5L1 | 222163_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.011
p=0.004
p=4e−04
p=0.0073
p=0.1049
p=0.071
p=0.3815
p=0.8049
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

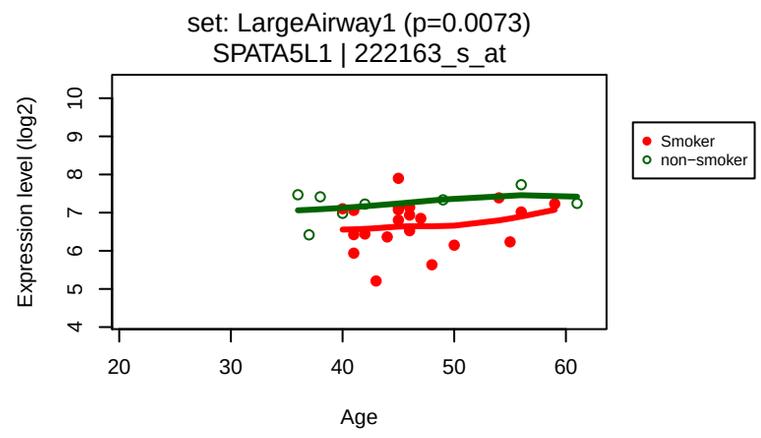

Gene expression ratio (smokers/nonsmokers)
gene: SPATA5L1 | 222163_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.011
p=0.004
p=4e−04
p=0.0073
p=0.1049
p=0.071
p=0.3815
p=0.8049
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

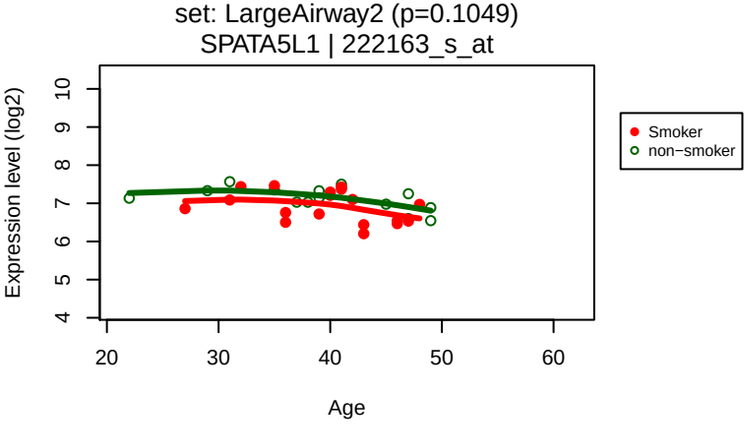

Gene expression ratio in Males
gene: SPATA5L1 | 222163_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0508
p=0.3558
p=0.4425
p=0.6958
SmallAirway2
LargeAirway2
Trachea
Macrophage2

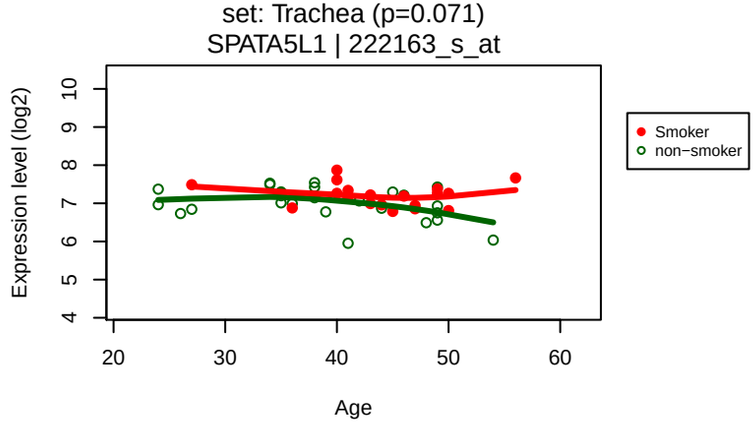

Gene expression ratio in Females
gene: SPATA5L1 | 222163_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0609
p=0.2837
p=0.0442
p=0.4407
SmallAirway2
LargeAirway2
Trachea
Macrophage2

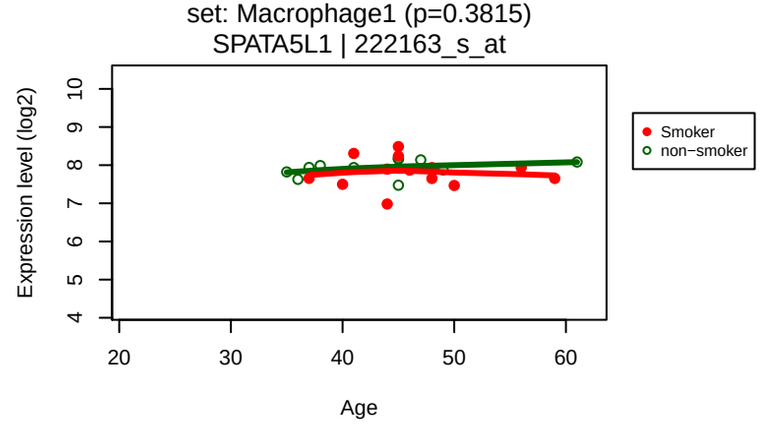

set: SmallAirway1 (p=0.011)
SPATA5L1 | 222163_s_at
Expression level (log2)
Age
Smoker
non−smoker

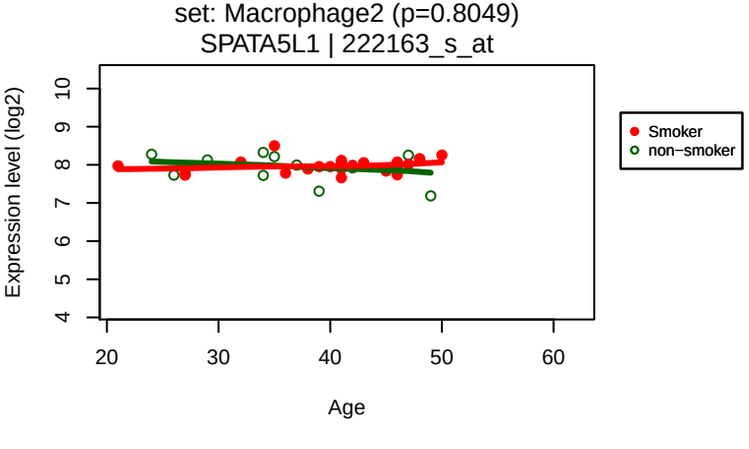

set: SmallAirway2 (p=0.004)
SPATA5L1 | 222163_s_at
Expression level (log2)
Age
Smoker
non−smoker

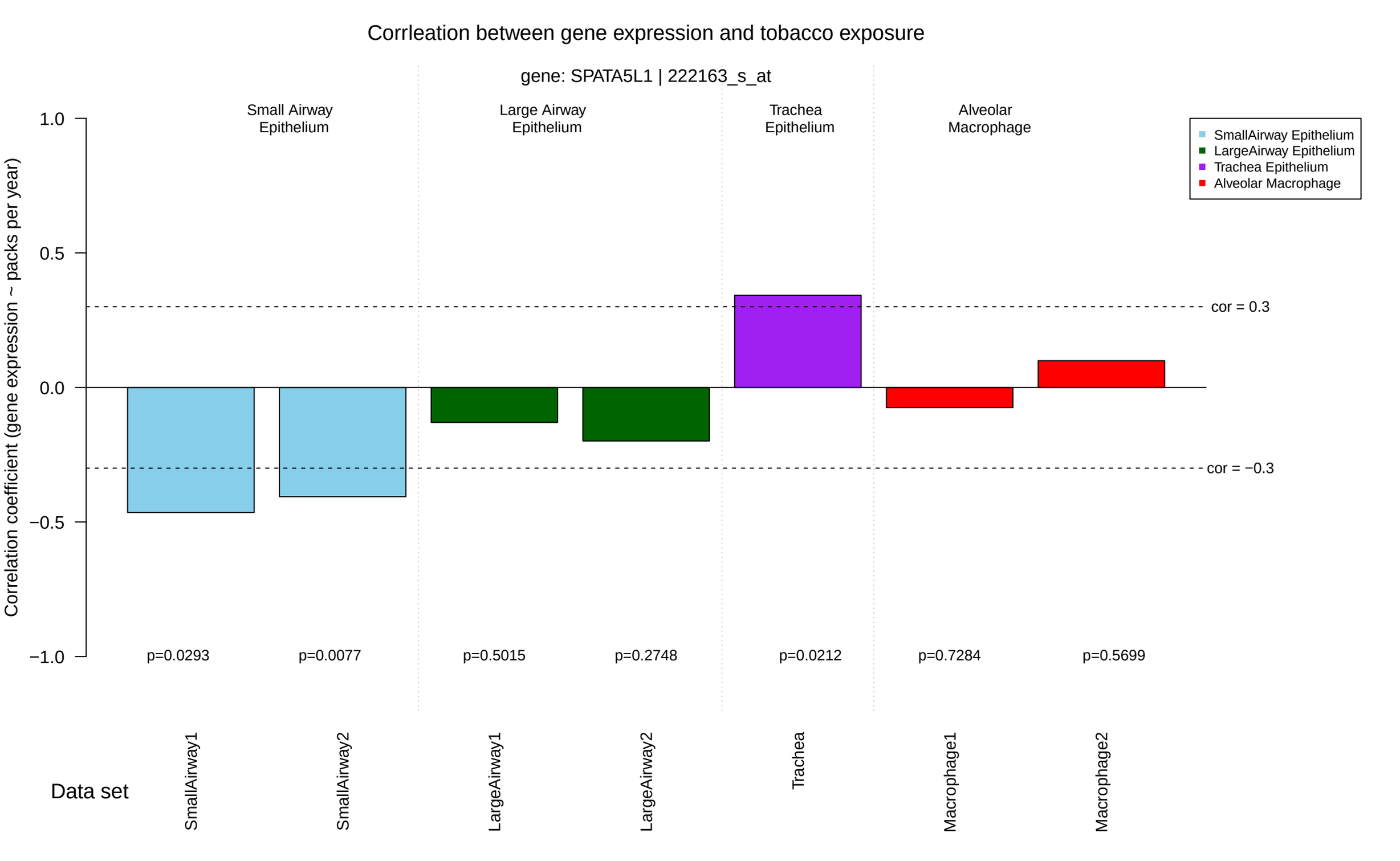

set: LargeAirway1 (p=0.0073)
SPATA5L1 | 222163_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1049)
SPATA5L1 | 222163_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.071)
SPATA5L1 | 222163_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3815)
SPATA5L1 | 222163_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8049)
SPATA5L1 | 222163_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA5L1 | 222163_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0293
p=0.0077
p=0.5015
p=0.2748
p=0.0212
p=0.7284
p=0.5699
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2